



UNITED INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT PROPOSAL

HaluRISC: Calibrated and Explainable Hallucination Risk Prediction for Large Language Model Outputs

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1 Project Overview

Large language models (LLMs) like ChatGPT are very good at writing text that sounds confident and correct, but they sometimes generate information that is wrong or made up — this is called a hallucination. This project builds a complete system that predicts how likely an LLM answer is to be hallucinated. The system is black-box friendly — it only needs the question, the answer, and (optionally) some context, without ever looking inside the LLM. It is lightweight — it uses simple text features and classical machine learning (XGBoost), so a full analysis runs in about 62ms at the median on a laptop, with no API fees. It is explainable — for every prediction it shows *why* the answer looks risky (for example: “the answer contradicts the provided context”, or “the answer contains numbers that are not supported”). Finally, it is presentable — a polished, interactive web dashboard lets a judge or supervisor test the system live, type any question and answer, and see the risk score and explanation instantly, with every claim in the answer verified against the pasted context, uploaded documents, or live web search, and cited. The novelty is not in inventing hallucination detection, but in combining NLI-based consistency features, calibrated risk scores, and tested explanations into one lightweight system — a combination no published work offers as a single product.

2 Motivation

LLMs are now used in real products — chatbots, Q&A systems, content review tools — where a wrong answer has a real cost, and the most common way to detect hallucinations is to ask the LLM itself to judge its own output [2], which is slow, expensive, and sometimes wrong. A lightweight model trained on simple text features (word overlap, named entities, contradiction with context, numbers, hedging words) can flag risky answers almost instantly and at almost zero cost. A risk score is also more useful than a simple “yes/no” flag, because real systems can decide what to do with a risky answer — show a warning, ask for human review, or fetch better evidence. And because the final result is a working product with a clean UI rather than just a report, the project is fully ready for live demonstration.

3 Objectives

Research question: Can a lightweight machine learning model predict hallucination risk in LLM answers using simple text features, while giving calibrated risk scores and clear explanations? To answer this, the project has five objectives. The first is to build a feature extraction pipeline (26 features in seven groups) that captures how well an answer agrees with the question and context, covering word overlap, named entities, contradiction signals, numbers, hedging words, and semantic similarity. The second is to train and honestly compare classical models, including a simple heuristic baseline, Logistic Regression, Random Forest, and XGBoost. The third is to calibrate the risk scores so that a score of 0.8 truly means “about 80% risk”, making the score trustworthy for real use. The fourth is to explain predictions with SHAP, giving a global view of which features matter most plus detailed case studies of individual predictions. The fifth is to build a polished web dashboard (Next.js 16 with assistant-ui + FastAPI) with strong UI/UX, featuring live prediction, a risk gauge, per-claim evidence verdicts, an explanation panel, and a demo gallery, designed for presentation and judging.

4 Related Work

Hallucination detection is an active research area. The most common approach asks the LLM itself to judge its own answer, as in SelfCheckGPT [2], but this is slow and expensive. To cut

this cost, small language models have been used as detectors [9], and lightweight evaluation models such as Luna [10] show that efficient detection is possible in practice. Other approaches train neural classifiers on frozen language encoders [7], or combine explainable text signals with machine learning [6, 8].

On the benchmark side, HaluEval [1] and TruthfulQA [3] provide standard test beds, and fine-grained methods such as FActScore [4] evaluate factual precision in detail. RAGTruth [13] adds natural responses with human word-level annotations. Newer benchmarks such as FaithBench [11] show that current detectors still fail on challenging cases, and SpikeScore [12] shows that cross-domain generalization remains an open problem.

None of these works combine lightweight engineered features with calibrated risk scores, tested explanations, and a deployable dashboard — this is the gap the project fills.

5 Methodology

The pipeline has seven clear steps, all implemented and verified. The first two prepare the data and the features. The QA subset of HaluEval is downloaded, cleaned, and manually inspected on 50 random samples to verify label quality, and each answer is described by 26 features in seven groups covering length and style, lexical overlap, entity overlap, NLI contradiction with a lightweight pretrained NLI model, numeric consistency, hedging phrases, and semantic similarity.

Model development comes next. A simple heuristic baseline, Logistic Regression, Random Forest, and XGBoost are compared with grouped 5-fold cross-validation, randomized hyperparameter search, and three-seed runs, judged on F1, AUROC, precision, recall, and MCC. Platt and isotonic calibration are fit on the validation split only and measured with expected calibration error and Brier score, SHAP provides global feature importance plus three local case studies, and the worst predictions are examined in a short error analysis.

The last step is deployment. A FastAPI backend serves the trained model, and a Next.js 16 + assistant-ui dashboard provides the risk gauge, the feature breakdown, per-claim evidence verdicts with citations, and a live input panel.

Figure 1 shows the end-to-end pipeline, and Figure 2 shows the deployed two-tier architecture.

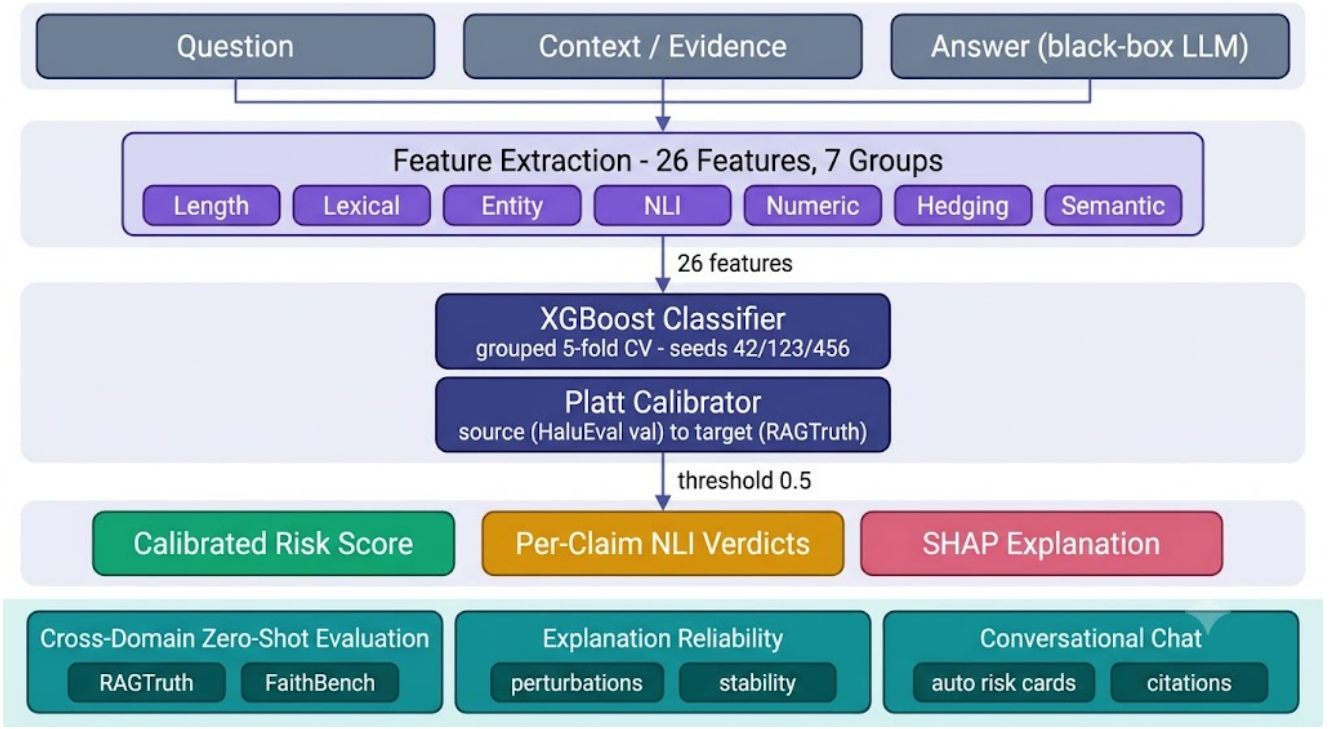


Figure 1: The HaluRISC pipeline: evidence-consistency features from the question, context, and answer feed a tuned, Platt-calibrated XGBoost model, producing a calibrated risk score, per-claim NLI verdicts, and SHAP explanations.

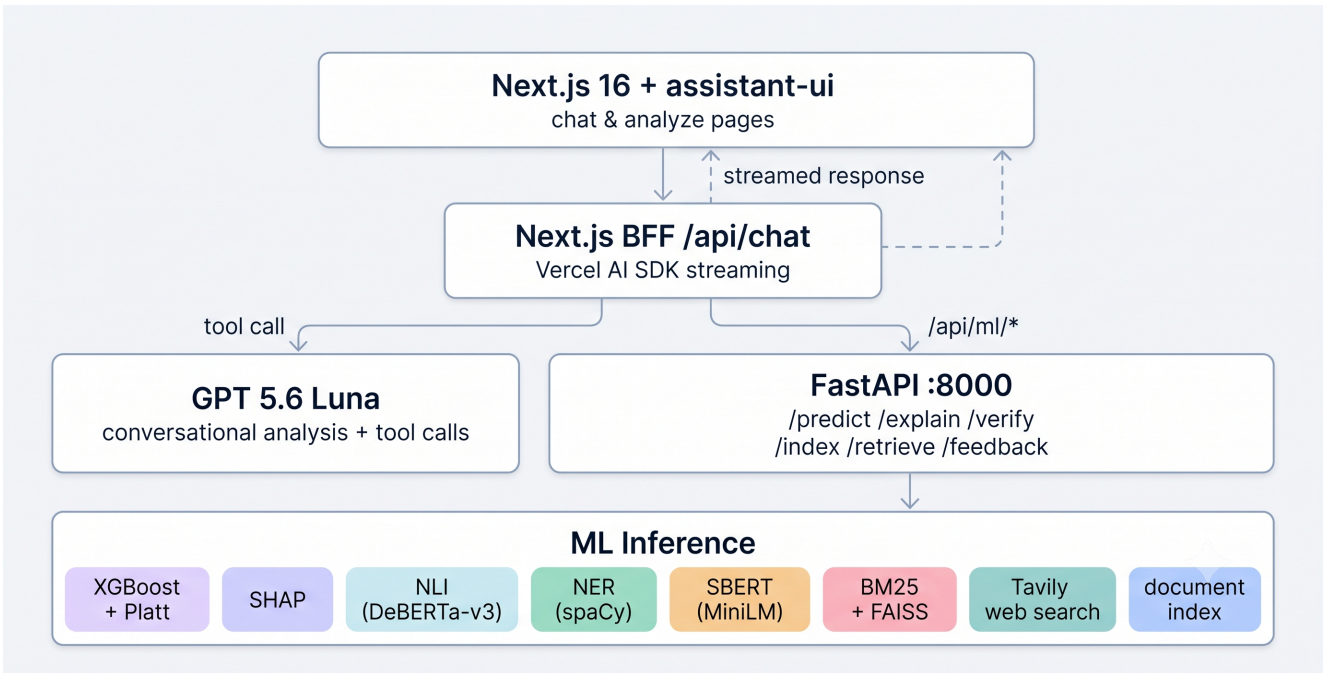


Figure 2: Deployed two-tier architecture: the Next.js frontend streams the conversation through a backend-for-frontend proxy to FastAPI, which serves the machine-learning endpoints and the document and live web-search evidence.

Statistical rigor: model differences are tested with McNemar’s test and bootstrap confidence intervals, so conclusions are not based on chance from a single split.

External comparison: results are compared with published HaluEval detection baselines, such as SelfCheckGPT [2], and validated on the RAGTruth corpus [13], so a judge can see whether the

scores are good and whether they generalize.

6 Dataset

The project uses three public corpora. The training source is the QA subset of HaluEval [1] (EMNLP 2023), the standard benchmark for hallucination evaluation. It contains 10,000 questions, each paired with a correct answer and a hallucinated answer, which yields 20,000 labeled rows with a balanced label distribution. The rows are split 70/15/15 at the question-group level into 14,000 training, 3,000 validation, and 3,000 test rows, so the two answers of one question never cross splits.

The evaluation corpora come from the extended study. RAGTruth [13] contributes 17,790 naturally generated responses with human word-level annotations, covering three tasks (QA, summarization, and data-to-text), four domains (MS MARCO, CNN/DailyMail, recent news, and Yelp), and six generator models from the GPT, Llama-2, and Mistral families. FaithBench [11] adds 750 summaries produced by ten modern LLMs. Both corpora are used for zero-shot evaluation, and a 5,034-row RAGTruth QA slice serves as the calibration set for target-domain recalibration, with a disjoint 900-row held-out test set.

7 Expected Results

The experiments are complete, and the expected results are confirmed on the frozen benchmark. On the HaluEval QA subset (20,000 rows, 70/15/15 grouped stratified split) the tuned XGBoost model reaches $F1 = 0.9846$, $AUROC = 0.9979$, and $MCC = 0.9697$, with an expected calibration error of 0.0045 before any recalibration. The full analysis runs in about 62ms at the median and costs about \$0.001 per 1,000 predictions, roughly 100 times cheaper than an LLM-as-judge that it outperforms on a measured 200-sample set ($F1$ 0.985 vs. 0.841). Zero-shot transfer to natural RAGTruth responses is poor, with $F1 = 0.3022$ on the QA split, an honest negative result reported rather than hidden, and a small target-domain recalibration on 5,034 RAGTruth rows cuts the ECE on that corpus from 0.8185 to 0.1335. The project delivers a trained and calibrated XGBoost model with feature ablation and SHAP explanations, statistical significance tests, a judge-ready web dashboard with per-claim evidence verdicts and citations, and fully reproducible code and artifacts.

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